CHAPTER 3

DYNAMICS

Key idea: a resultant force changes momentum; in an isolated system the total momentum is conserved.

1 NEWTON'S LAWS

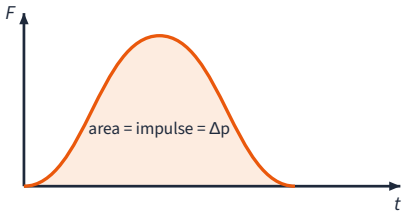
1. **1st:** a body stays at rest or at constant velocity unless a resultant force acts (inertia, measured by mass).
2. **2nd:** rate of change of momentum ∝ resultant force, in its direction.
3. **3rd:** A on B = − B on A; the pair act on **different** bodies.

$$F = \frac{dp}{dt} \qquad F = ma$$

1 N gives 1 kg an acceleration of 1 m s^{-2}.

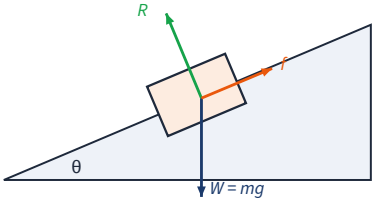
2 MOMENTUM & IMPULSE

$$p = mv \qquad J = F\Delta t = \Delta p$$



- Impulse = **area** under the F–t graph
- Unit: kg m s^{-1} = N s

3 FREE-BODY DIAGRAMS



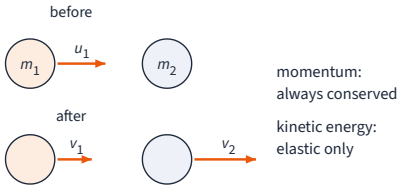
- Smooth incline: $a = g\sin\theta$, $N = mg\cos\theta$
- Lift up: $R = m(g + a)$; down: $R = m(g - a)$
- Free fall: $R = 0$ (weightless)

4 CONSERVATION OF MOMENTUM

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

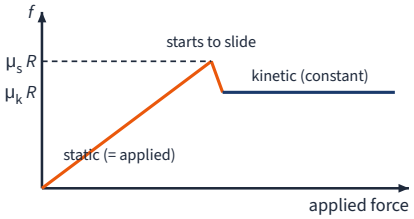
- Isolated system: no resultant external force
- Holds in **every** collision and explosion
- Recoil from rest: $m_1v_1 + m_2v_2 = 0$
- 2-D: conserve p_x and p_y separately

5 COLLISIONS



Type	Momentum	Kinetic energy
Elastic	conserved	conserved
Inelastic	conserved	lost
Perfectly inelastic	conserved	max. loss, stick

6 FRICTION



$$f_s \le \mu_s N \qquad f_k = \mu_k N$$

- Opposes (the tendency of) relative motion
- $\mu_k < \mu_s$; independent of contact area
- About to slide on incline: $\tan\theta = \mu_s$

7 USEFUL RESULTS

Elastic, m_2 at rest:

$$v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2} \qquad v_2 = \frac{2m_1u_1}{m_1 + m_2}$$

Elastic: speed of approach = speed of separation

$$u_1 - u_2 = v_2 - v_1$$

Kinetic energy: $E_k = \dfrac{p^2}{2m}$

Pulley (light string): $a = \dfrac{(m_1 - m_2)g}{m_1 + m_2}$, $T = \dfrac{2m_1m_2g}{m_1 + m_2}$. Stream on a wall: $F = \dfrac{dm}{dt}(v - u)$.

8 COMMON SECTION A TRAPS

- Calling weight and normal reaction a 3rd-law pair (same body, different types).
- Rebound: $\Delta p = m(v + u)$, not $m(v - u)$.
- Forgetting momentum is a vector: fix a positive direction.
- Assuming $N = mg$ when the force pulls at an angle.

EXAM TIPS

- ✓Draw a free-body diagram for each body first.
- ✓Apply $\Sigma F = ma$ along the direction of motion.
- ✓State "no external force, so momentum is conserved".
- ✓Check kinetic energy to decide whether a collision is elastic.